# ANOOP SENTHIL

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## Summary

AI/ML Engineer with an M.Sc in Artificial Intelligence and hands-on experience across LLM engineering, machine learning, computer vision, and AI systems. Experienced in designing, training, and evaluating intelligent systems, with expertise spanning LLMs, RAG, multimodal models, model fine-tuning, and scalable software development. Interested in building reliable AI systems that translate advanced ML research into practical, production-ready applications.

## Education

**Master of Science, Artificial Intelligence** **Sep 2025 - Sep 2026**
*Queen Mary University of London* *London, United Kingdom*
- **Achievements:** Recipient of the Global Talent Scholarship (£5,000) for academic excellence
- **Coursework:** Machine Learning, Neural Networks, NLP, Information Retrieval, AI Ethics, Reinforcement Learning

**Bachelor of Technology, Computer Science and Engineering** **Jul 2021 - Jul 2025**
*Vellore Institute of Technology* *Vellore, India*
- **Coursework:** Data Structures & Algorithms, Database Management, Cloud Computing, Software Engineering

## Projects

**Hybrid RAG System for Scientific Question Answering** | GitHub **Aug 2026 - Sep 2026**
- Built a hybrid retrieval-augmented generation pipeline over the SciFact corpus, combining sparse BM25 first-stage retrieval with ColBERT-style late-interaction re-ranking to supply grounded, citation-backed answers from a 4-bit quantized Qwen2.5-3B large language model.
- Implemented a two-stage pipeline where BM25 surfaces top-100 candidates for ColBERT MaxSim re-ranking before the top-5 passages go to Qwen for cited generation, with the ColBERT projection layer fine-tuned via InfoNCE on BM25-mined hard negatives to sharpen domain-specific relevance ranking.
- Evaluated hybrid retrieval on 300 test queries, achieving 0.6858 nDCG@10, 0.6479 AP@10, and 0.6581 RR@10, with a re-ranking latency of 43.1ms/query.

**Autonomous Equity Research & Valuation Agent** | GitHub **Apr 2026 - Sep 2026**
- Built a multi-agent financial research application with CrewAI and Groq, orchestrating natural-language ticker queries across Python-based market data, RSS news, SEC filings, and automated ReportLab PDF generation.
- Developed AI agents for document Q&A, bull/bear analysis, sentiment and risk assessment, alongside a 10-year DCF valuation model using WACC and rule-based stock recommendations.
- Built a Qdrant-based RAG pipeline over SEC 10-K Item 1A disclosures using MiniLM embeddings and containerized the application with Docker Compose, with GPT-OSS 20B LLM inference served through Groq.

**Evolutionary-Augmented Image-Text Retrieval System** | GitHub **Oct 2025 - Nov 2026**
- Designed a lightweight multimodal search system retrieving semantically relevant text labels for any input image, enabling accurate visual search without large pre-trained foundation models or expensive retraining.
- Constructed a custom Skip-Gram language backbone trained on a Visual Genome word co-occurrence knowledge graph stored in Neo4j, then expanded it via a (1+λ) evolutionary algorithm to integrate 68 new CIFAR-100 labels while preserving its semantic structure without retraining.
- Aligned MobileNetV3 image features to the enriched text space using InfoNCE contrastive loss, achieving 91% Recall@10 for image-to-text retrieval on the CIFAR-100 test set, with strong generalization across related categories.

## Preprints

**ReVA: A Region-Aware Visual Assistant for Visually Grounded Question Answering** | arXiv | GitHub
- Developed ReVA to reduce object hallucination in visual question answering by explicitly grounding language generation in both global scene context and localized visual evidence, coupling CLIP ViT-L/14 Vision Transformer (ViT) with Qwen2.5-7B-Instruct LLM through separate whole-image and regional projection bridges.
- To provide the LLM with object-specific evidence for questions requiring fine-grained visual understanding, used RAM++ to identify question-agnostic objects present in the image and spaCy to extract question-dependent objects referenced by the question, Grounding DINO then localized these objects with bounding boxes, enabling RoI Align to extract their visual features from intermediate ViT representations.
- To integrate information across visual abstraction levels without discarding either low-level detail or high-level semantics, used cross-depth self-attention and multi-token pooling, followed by joint captioning and InfoNCE training and LoRA fine-tuning of the LLM to align visual representations with language.
- Evaluated ReVA on VQAv2, MMBench, POPE, and SEED-Bench benchmarks, achieving 73.79% VQAv2 accuracy and 82.85% mean F1 on POPE, while improving MMBench localization by 8.65 points and attribute comparison by 6.82 points.

## Experience

**DRDO, Ministry of Defence, Govt. of India** **Jan 2025 - May 2025**
*AI Research and Development Intern* *Bangalore, India*
- Developed an end-to-end image-to-code automation pipeline utilizing Deep Learning and Computer Vision to interpret control-law diagrams and synthesize structured C code, completely eliminating manual human translation errors.
- Fine-tuned a YOLO object detector to identify block components and designed a robust OpenCV line extraction pipeline incorporating CLAHE for contrast enhancement, Bilateral Filtering for noise reduction, and Hough Transform to trace line connections within noisy diagrams for downstream graph construction.
- Built an autonomous pipeline that processed images, translated them into directed acyclic graph (DAG) representations, performed graph reasoning for logical code generation, and ran GCC syntax validation, which verified the correctness of the generated C code before execution.

**Pivotrics** **Aug 2023 - Nov 2023**
*Software Developer Intern* *Bangalore, India*
- Developed scalable RESTful APIs for a SaaS payment platform using Spring Boot, WebFlux, and PostgreSQL, with non-blocking async APIs and back-pressure management for data stream processing.
- Secured product and customer data by implementing JWT-based authentication and role-based access control using Spring Security.

## Skills

- **Languages**: Python, Java, C, HTML/CSS, JavaScript, SQL, Cypher
- **Frameworks & Libraries**: PyTorch, HuggingFace, LangChain, LangGraph, CrewAI, OpenCV, spaCy, Spring Boot
- **Developer Tools**: Cursor, Claude Code, Git, GitHub, Jupyter, Docker, BMAD Method, Spec Kit
- **Databases & Cloud**: PostgreSQL, Qdrant, Neo4j, Amazon Web Services